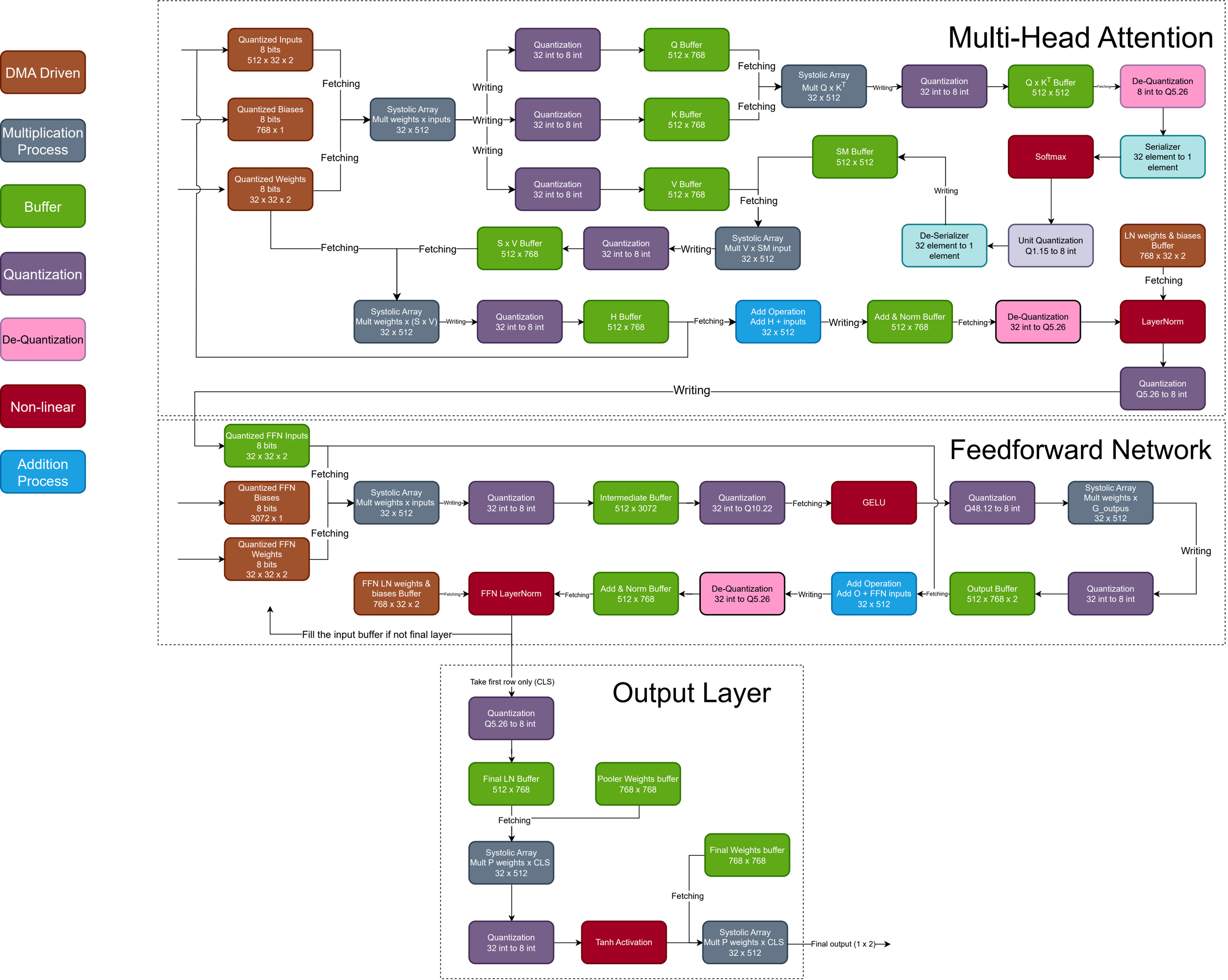
DMA Driven
Multiplication Process
Buffer
Quantization
De-Quantization
Non-linear
Addition Process
Multi-Head Attention
Quantized Inputs 8 bits 512 x 32 x 2
Quantized Biases 8 bits 768 x 1
Quantized Weights 8 bits 32 x 32 x 2
Fetching
Systolic Array Mult weights x inputs 32 x 512
Writing
Quantization 32 int to 8 int
Q Buffer 512 x 768
K Buffer 512 x 768
V Buffer 512 x 768
Systolic Array Mult Q x K^T 32 x 512
Q x K^T Buffer 512 x 512
De-Quantization 8 int to Q5.26
Serializer 32 element to 1 element
Softmax
Unit Quantization Q1.15 to 8 int
De-Serializer 32 element to 1 element
SM Buffer 512 x 512
Systolic Array Mult V x SM input 32 x 512
S x V Buffer 512 x 768
Systolic Array Mult weights x (S x V) 32 x 512
H Buffer 512 x 768
Add Operation Add H + inputs 32 x 512
Add & Norm Buffer 512 x 768
LN weights & biases Buffer 768 x 32 x 2
LayerNorm
Quantization Q5.26 to 8 int
Feedforward Network
Quantized FFN Inputs 8 bits 32 x 32 x 2
Quantized FFN Biases 8 bits 3072 x 1
Quantized FFN Weights 8 bits 32 x 32 x 2
Intermediate Buffer 512 x 3072
Quantization 32 int to Q10.22
GELU
Quantization Q48.12 to 8 int
Systolic Array Mult weights x G_outpus 32 x 512
Output Buffer 512 x 768 x 2
Add Operation Add O + FFN inputs 32 x 512
De-Quantization 32 int to Q5.26
Add & Norm Buffer 512 x 768
FFN LN weights & biases Buffer 768 x 32 x 2
FFN LayerNorm
Fill the input buffer if not final layer
Output Layer
Take first row only (CLS)
Final LN Buffer 512 x 768
Pooler Weights buffer 768 x 768
Systolic Array Mult P weights x CLS 32 x 512
Final Weights buffer 768 x 768
Tanh Activation
Final output (1 x 2)